

Exercise 9.1: The OOD Problem Your CARLA model was trained on sunny daytime images. At deployment, it may encounter images from fog or night - conditions outside its ODD.

1. Why can a standard classifier not be trusted to signal when it receives an OOD input?

- A standard classifier is trained only to classify known training data.
- It is not trained to recognize unfamiliar inputs.
- Therefore, it can produce high-confidence predictions even for OOD images.
- The model cannot reliably say "I don't know."

Example

A model trained only on sunny images may classify a night image as "No Pedestrian" with 98% confidence even though it has never seen night conditions.

2. Why is silent failure (confidently wrong prediction) worse than uncertain failure in a safety-critical system?

- Silent failure is dangerous because the system is confidently wrong.
- The system has no warning that the prediction may be incorrect.
- Uncertain failure is safer because the system knows it is unsure and can take precautionary actions.
- In safety-critical systems, detecting uncertainty helps prevent accidents.

Example

A self-driving car confidently misses a pedestrian in a night image and fails to brake, whereas an uncertain prediction could trigger a safety response.

Exercise 9.2: Baseline OOD Detection Describe the maximum softmax probability (MSP) baseline for OOD detection. How does it work, and what are its main limitations?

- Maximum Softmax Probability (MSP) is a simple baseline method for OOD detection.
- It uses the highest predicted probability from the classifier as the confidence score.
- Images with high confidence are considered in-distribution, while images with low confidence are considered OOD.
- The method is easy to implement and does not require modifying the model.

Limitation:

- Neural networks can produce high-confidence predictions for OOD inputs.
- Therefore, MSP may fail to detect some OOD samples and can result in confident but incorrect predictions.

Example: A model trained on sunny images may classify a night image with 99% confidence even though it has never seen nighttime conditions.

MSP assumes that unfamiliar inputs produce low confidence, but neural networks can still be confidently wrong.

Exercise 9.3: Alternative Methods Describe one more advanced OOD detection method (e.g., energy score or Mahalanobis distance). How does it improve over MSP?

- Mahalanobis Distance is a feature-based OOD detection method.
- It uses deep features extracted from the neural network instead of prediction confidence.
- The distance between a test image and the distribution of training features is computed.
- Images with large distances are considered OOD.

Improvement over MSP:

- MSP relies only on prediction confidence.
- Mahalanobis uses information from the learned feature space.
- It can detect OOD inputs even when the classifier is highly confident.
- Therefore, it is often more reliable than MSP for OOD detection.

Example:

A night image may receive a high-confidence prediction from the classifier, but its feature representation can still be far from the training distribution, allowing Mahalanobis distance to identify it as OOD.

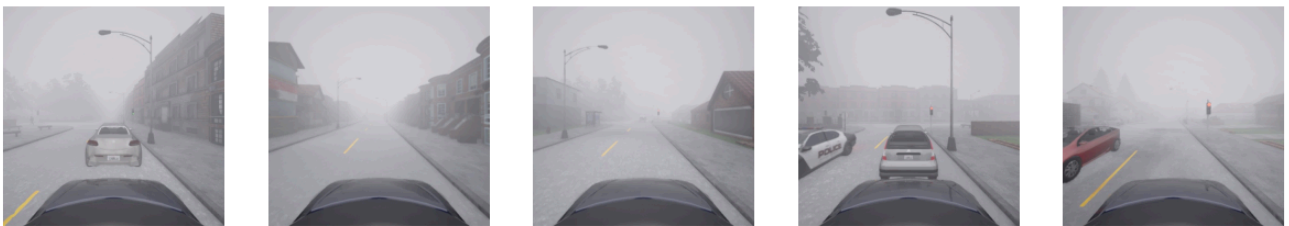
Exercise 9.4: Visualising the Distribution Shift

1. Display five in-distribution (sunny / daytime) images next to five fog and five night images.

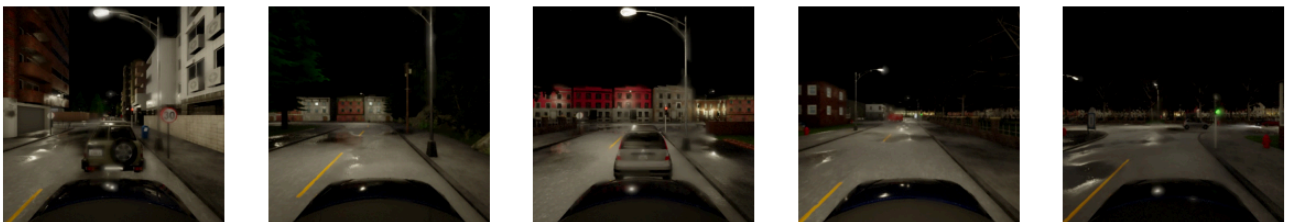
Sunny Images



Fog Images



Night Images



2. Also display five images from the different CARLA town. How do they differ from your training images, and how does that difference compare to the fog / night shift?

Different Town Images



- The fog images have reduced visibility and contrast compared to the training images.
- Night images differ significantly due to low illumination and strong lighting effects.
- The different-town images retain the same daytime and weather conditions as the training data but contain different buildings, road layouts, and vegetation.
- Therefore, the fog and night shifts are stronger distribution shifts than the different-town images.

3. Compute the mean softmax confidence of each of your three models on in-distribution vs. fog/night images. Is the model more or less confident on these inputs?

Model	Sunny	Fog	Night
Pedestrian	0.904	0.990	0.972
Vehicle	0.910	0.820	0.764
Traffic Light	0.959	0.822	0.885

Pedestrian Model

- Confidence increased on fog and night images.
- The model remained highly confident even on OOD data.
- This suggests possible overconfidence and highlights a limitation of MSP.

Vehicle Model

- Confidence decreased in fog and night conditions.
- The largest drop occurred at night.
- The model became less certain when encountering unfamiliar conditions.

Traffic Light Model

- Confidence decreased in fog and night images.
- The reduction was smaller than for the vehicle model.
- The model was still reasonably confident at night.

Exercise 9.5: Is the Different Town Out-of-Distribution? The different-town images were taken under nominal weather and lighting - the kind of conditions your system is meant to handle - but in a location your models never saw during training.

1. Look back at the ODD you specified in Exercise 2.2. As written, does your ODD clearly decide whether the different-town images are inside or outside it?

OOD is defined by the ODD, not by whether the model has seen the exact data before.

- No, the ODD does not clearly specify whether different-town images are inside or outside the ODD.
- The ODD only mentions urban roads and intersections, but it does not mention specific towns.
- Therefore, the status of the different-town images is ambiguous.

Example: The new town contains urban roads, but the ODD does not explicitly state whether unseen towns are allowed.

2. Revise the wording of your ODD so that the new town is unambiguously either inside or outside it. State which you chose and why ?

- Revised ODD: Urban roads and intersections under clear daytime conditions, regardless of the specific CARLA town.
- I consider the different-town images to be inside the ODD.
- The weather, lighting, and driving environment remain within the intended operating conditions.
- A change in town layout should be treated as normal variation that the system should handle.

Example: Different buildings and road layouts should not prevent an autonomous vehicle from operating safely.

3. What does that choice imply for how the town images should be treated - as inputs the system must handle correctly, or as inputs an OOD monitor should flag?

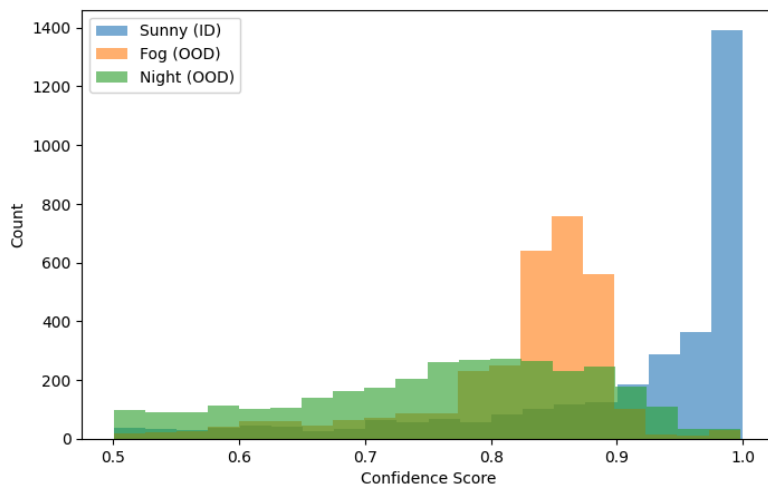
- The different-town images should be treated as inputs the system must handle correctly.
- They should not be flagged as OOD inputs.
- If the model performs poorly on these images, it indicates a generalization problem rather than an OOD problem.

Example: If the model fails in a new town, the solution is to improve the model's robustness rather than reject the input as OOD.

Exercise 9.6: Evaluating the MSP Baseline

Pick any one of your three models and use its maximum softmax probability as an OOD score (low confidence $\rightarrow$ OOD). Treat the sunny/daytime test images as in-distribution and the provided OOD images as OOD.

1. Plot the distribution of OOD scores for in-distribution and OOD images.



- The **Sunny (ID)** images are concentrated at very high confidence values (around 0.95–1.0).
- The **Fog (OOD)** images have lower confidence and are mostly between 0.8–0.9.
- The **Night (OOD)** images are spread across a wider range and have the lowest confidence overall.
- There is some overlap between the distributions, which means MSP cannot perfectly separate ID and OOD images.

2. Compute the AUROC for separating in-distribution from OOD, over all of the OOD scenarios.

AUROC: 0.8168

- The MSP baseline can distinguish ID and OOD images reasonably well.
- The separation is not perfect because some OOD images still receive high confidence scores.
- Overall, the detector performs better than random guessing and provides useful OOD detection capability.

Exercise 9.7: Feature-Based OOD Detection Implement one feature-based detector (Mahalanobis distance or k-NN, both covered in the lecture) and compare it to the MSP baseline.

1. Extract deep features for your in-distribution training (or validation) images and for the test in-distribution and OOD images. Use the same model whose MSP you evaluated in Exercise 9.6.

2. Fit the detector on the in-distribution features only, then score the test images (larger distance → more OOD).

- The k-NN detector achieved an AUROC of 0.7799.
- The MSP baseline achieved an AUROC of 0.8168.
- In this experiment, MSP performed slightly better than the feature-based detector.

3. Compute its AUROC and compare it to the MSP AUROC from Exercise 9.6. For which OOD scenario is the gap largest?

- Fog AUROC: 0.5973
- Night AUROC: 0.9625

Observation

- The detector performed very well on night images.
- Fog images were much harder to distinguish from in-distribution images.
- The largest gap was observed between the fog and night scenarios.

Reason

- Night images differ significantly from the training distribution in terms of lighting.
- Fog images still contain similar roads, buildings, and scene structures, making them harder to detect as OOD.

The k-NN feature-based detector was highly effective for night images but struggled with fog images, while MSP achieved slightly better overall OOD detection performance.

Exercise 9.8: Extending the Safety Analysis for OOD You have now seen how easily the perception models fail - silently - on out-of-ODD inputs, and how well (or poorly) an OOD monitor can catch them. Revisit your System-Theoretic Process Analysis from Sheet 2 and extend it to cover this risk. Many of you will not yet have an OOD-related entry in your analysis; the goal of this exercise is to add one and trace it through.

1. Hazard. Check the hazards from Exercise 2.4. Is operating on an undetected outof-ODD input captured (e.g. as a cause of “vehicle fails to brake for a pedestrian”)? If your hazard list does not reflect it, refine or add a hazard.

Exercise 2.4: Hazards Updated:

- H-4 already considers operation outside the ODD.
- However, it does not explicitly cover undetected OOD inputs.
- New Hazard (H-5): The system relies on perception outputs from undetected out-of-ODD inputs, resulting in incorrect detections and unsafe vehicle behavior.

Example: A night image is not detected as OOD, causing the model to miss a pedestrian.

ID	Hazard	Loss(es)	Likelihood	Severity
H-1	Vehicle keeps moving when a pedestrian is in front	L-1	Medium	High
H-2	System fails to detect an obstacle (pedestrian/vehicle)	L-1, L-2	Medium	High
H-3	Vehicle does not obey traffic signals (e.g., runs a red light)	L-3, L-1	Low	High
H-4	System operates outside its ODD (e.g., night or bad weather)	L-1, L-2, L-3	Medium	High
H-5	System relies on perception outputs from undetected out-of-ODD inputs, resulting in incorrect detections and unsafe vehicle behavior	L-1, L-2, L-3	Medium	High

2. Unsafe control action. Review the UCAs from Exercise 2.6. Add a UCA for the case where the planner acts on perception output that is unreliable because the input is out-of-distribution and this is not detected (e.g. “the planner continues at speed while the camera input is out-of-ODD and the perception output is untrustworthy”). Link it to the hazard(s) above.

UCA-9

- Controller: Planning Module
- Control Action: Continue Driving
- UCA Type: Provided Unsafely
- Hazard: H-5
- Unsafe Scenario: The planner continues normal driving while the camera input is outside the ODD and the unreliable perception output is trusted.

Updated Ex: 2.6:

ID	Controller	Control Action	UCA Type	Hazard(s)	Unsafe Scenario
UCA-1	Planning Module	Brake	Not provided	H-1	Vehicle does not brake when a pedestrian is in front
UCA-2	Planning Module	Brake	Provided unsafely	H-2	Vehicle brakes even when there is no obstacle
UCA-3	Planning Module	Brake	Wrong timing	H-1	Vehicle brakes too late after detecting a pedestrian
UCA-4	Planning Module	Brake	Wrong duration	H-1	Vehicle stops braking too early while pedestrian is still present
UCA-5	Human Operator	Take control	Not provided	H-4	Human does not take control when system is outside ODD
UCA-6	Human Operator	Take control	Provided unsafely	H-2	Human interferes unnecessarily when system is working correctly
UCA-7	Human Operator	Take control	Wrong timing	H-1	Human reacts too late to avoid a collision
UCA-8	Human Operator	Take control	Wrong duration	H-3	Human takes control briefly and releases it too early
UCA-9	Planning Module	Continue Driving	Provided Unsafely	H-5	The planner continues normal driving while the camera input is outside the ODD and the unreliable perception output is trusted.

3. Safety constraints. From that UCA, derive at least one new safety constraint of each kind (cf. Exercise 2.7):

- a model-level constraint on the OOD monitor, justified by the severity of the hazard it guards against; and
- a system-level constraint on the response.

Model-Level Constraint

- The OOD monitor must detect and flag inputs that significantly differ from the training distribution.
- Justification: Failure to detect OOD inputs may lead to incorrect perception and severe safety consequences.

System-Level Constraint

- When an OOD input is detected, the vehicle must enter a safe state by slowing down or requesting human intervention.
- Justification: Safe behavior reduces the risk of accidents caused by unreliable perception outputs.

Updated exercise 2.7: Safety Constraints :

UCA	Safety Constraint	Level	Verification
UCA-1	The system must apply braking when a pedestrian is detected in front	System-level	Test in simulation scenarios with pedestrians
UCA-2	The system should not brake unless a valid obstacle is detected	Model-level	Evaluate false positive rate of perception model
UCA-3	The system must respond within a safe time after detecting an obstacle	System-level	Measure response latency in tests
UCA-4	The system must continue braking until the path is clear	System-level	Simulation testing with moving pedestrians
UCA-5	The human operator must take control when the system operates outside ODD	System-level	Check system alerts and human response
UCA-6	The system should avoid unnecessary human intervention during normal operation	System-level	Monitor system behavior in safe conditions
UCA-7	The human operator should react within a safe time when intervention is required	System-level	Measure reaction time in test scenarios
UCA-8	The human operator must maintain control until the situation is safe	System-level	Evaluate behavior during takeover scenarios
UCA-9A	The OOD monitor should achieve sufficient detection performance on representative OOD scenarios (e.g., fog and night) and trigger a warning when confidence in the perception system is low.	Model-level	Evaluate AUROC of the OOD detector on fog and night datasets.
UCA-9B	When an OOD input is detected, the vehicle must enter a safe state by slowing down or requesting human intervention.	System-level	Test vehicle response in simulated OOD scenarios.

4. Residual risk. Even with a perfect OOD detector, what risk remains? Does detecting OOD fully address the UCA and hazard?

- Even with an OOD detector, some OOD inputs may not be detected correctly.
- In our experiments, fog images were significantly harder to detect than night images (AUROC 0.5973 vs. 0.9625).
- Therefore, unsafe situations may still occur if the OOD monitor fails to identify certain distribution shifts.
- OOD detection reduces risk but does not completely eliminate hazards caused by unreliable perception outputs.

Updated Exercise 2.8: Causal Loss Scenarios

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1	Camera fails to detect a pedestrian, so the system thinks the road is clear and does not brake	Poor detection or unclear image	System must brake when pedestrian is detected
UCA-2	System detects a false obstacle and brakes unnecessarily	Noise or incorrect model prediction	System should not brake without valid obstacle
UCA-3	Pedestrian is detected late, so braking happens too late	Processing delay or slow response	System must respond within safe time
UCA-4	System starts braking but stops too early while pedestrian is still present	Incorrect control logic	System must continue braking until safe
UCA-5	System operates outside ODD, but human does not take control	Human not attentive or no alert	Human must take control outside ODD
UCA-6	Human takes control even when not needed and causes disturbance	Lack of trust or confusion	Avoid unnecessary human intervention
UCA-7	Human reacts too late to take control in a risky situation	Slow reaction or distraction	Human must respond within safe time
UCA-8	Human takes control but releases it too early before situation is safe	Misjudgment of situation	Human must maintain control until safe
UCA-9	The camera input is outside the ODD, but the OOD monitor fails to detect it. The planner trusts the perception output and continues driving.	OOD detection failure or overconfident model prediction	OOD monitor must detect OOD inputs and trigger a safe response.